

Chapter 7

Efficient Diversification

INVESTMENTS

THIRTEENTH EDITION

BODIE, KANE, MARCUS

Chapter Overview

- The investment decision is a top-down process.
 1. Capital allocation (risky vs. risk-free).
 2. Asset allocation.
 3. Security selection.
- Optimal risky portfolio construction.
- Efficient diversification.
- Long-term versus short-term investment horizons.

Diversification and Portfolio Risk

- **Market risk.**
 - Attributable to market-wide risk sources.
 - Remains even after diversification.
 - Also called **systematic risk** or **nondiversifiable risk**.
- **Firm-specific risk.**
 - Risk that can be eliminated by diversification.
 - Also called **unique risk, diversifiable risk, or nonsystematic risk**.

Figure 7.1 Portfolio Risk as a Function of the Number of Stocks in the Portfolio

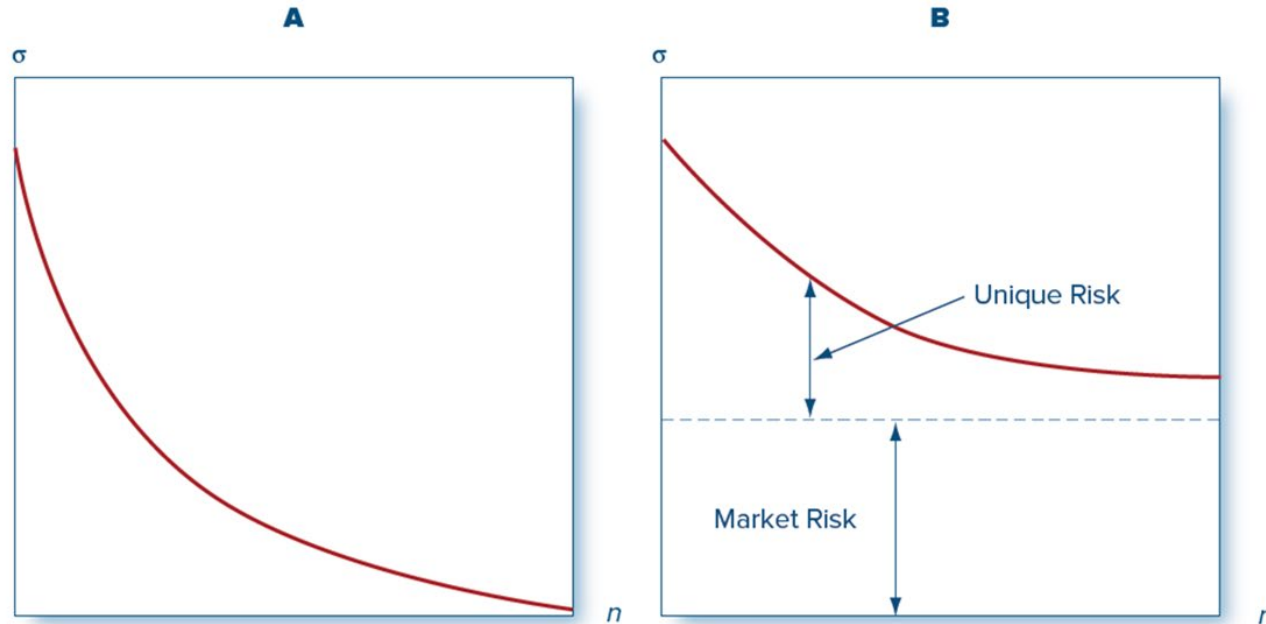


Figure 7.1 Portfolio risk as a function of the number of stocks in the portfolio.
Panel A: All risk is firm-specific. **Panel B:** Some risk is systematic or market-wide.

Insurance Principle: Risk reduction by spreading exposure across many independent risk sources.

Figure 7.2 Portfolio Diversification

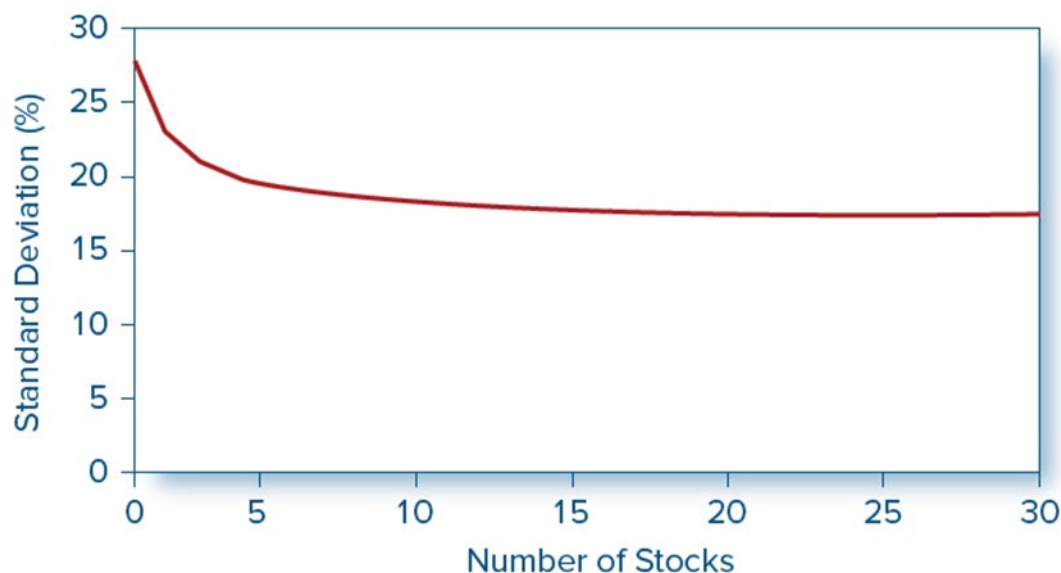


Figure 7.2 Portfolio diversification. The average annualized standard deviation of returns of portfolios composed of only one stock selected at random from the NYSE was 27.8% in the decade ending 2017. Portfolio risk fell rapidly as the number of stocks included in the portfolio increased. In the limit, portfolio risk could be reduced to only 17.4%.

Source: Authors' calculations. This is an update of a figure that first appeared in Statman, M. (1987, September). How many stocks make a diversified portfolio? *Journal of Financial and Quantitative Analysis*, 22.

Portfolios of Two Risky Assets

- Expected return.
 - Weighted average of expected returns on the component securities.
- Portfolio risk.
 - Variance of the portfolio is a weighted sum of covariances, and each weight is the product of the portfolio proportions of the pair of assets.

Portfolios of Two Risky Assets: Expected Return

Consider a portfolio made up of equity (stocks) and debt (bonds)...

$$r_p = w_D r_D + w_E r_E$$

Where:

r_p = rate of return on portfolio

w_D = proportion invested in the bond fund

w_E = proportion invested in the stock fund

r_D = rate of return on the debt fund

r_E = rate of return on the equity fund

$$E(r_p) = w_D E(r_D) + w_E E(r_E)$$

Portfolios of Two Risky Assets: Risk

Variance of r_P : (Handout for details)

$$\sigma_P^2 = w_D^2 \sigma_D^2 + w_E^2 \sigma_E^2 + 2w_D w_E \text{Cov}(r_D, r_E)$$

σ_D^2 = Bond Variance

σ_E^2 = Equity Variance

$\text{Cov}(r_D, r_E)$ = Covariance of returns for bonds and equity

Portfolios of Two Risky Assets: Covariance

Covariance of returns on bonds and equity:

$$Cov(r_D, r_E) = \rho_{DE} \sigma_D \sigma_E$$

ρ_{DE} = Correlation coefficient of returns

σ_D = Standard deviation of bond returns

σ_E = Standard deviation of equity returns

Portfolios of Two Risky Assets: Correlation Coefficients ₁

Range of values for correlation coefficient:

$$-1.0 \leq \rho \leq 1.0$$

- If $\rho = 1.0 \rightarrow$ perfectly positively correlated securities.
- If $\rho = 0 \rightarrow$ the securities are uncorrelated.
- If $\rho = -1.0 \rightarrow$ perfectly negatively correlated securities.

Portfolios of Two Risky Assets: Correlation Coefficients ²

When $\rho_{DE} = 1$, there is no diversification:

$$\sigma_P = w_E \sigma_E + w_D \sigma_D$$

When $\rho_{DE} = -1$, a perfect hedge is possible.

$$w_E = \frac{\sigma_D}{\sigma_D + \sigma_E} = 1 - w_D$$

Portfolios of Two Risky Assets: Example – 50%/50% Split

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Table 7.1

Descriptive statistics
for two mutual funds

	Debt	Equity
Expected return, $E(r)$	8%	13%
Standard deviation, σ	12%	20%
Covariance, $\text{Cov}(r_D, r_E)$	72	
Correlation coefficient, ρ_{DE}	0.30	

**Expected
Return:**

$$\begin{aligned}
 E(r_p) &= w_D E(r_D) + w_E E(r_E) \\
 &= .50 \times 8\% + .50 \times 13\% = 10.5\%
 \end{aligned}$$

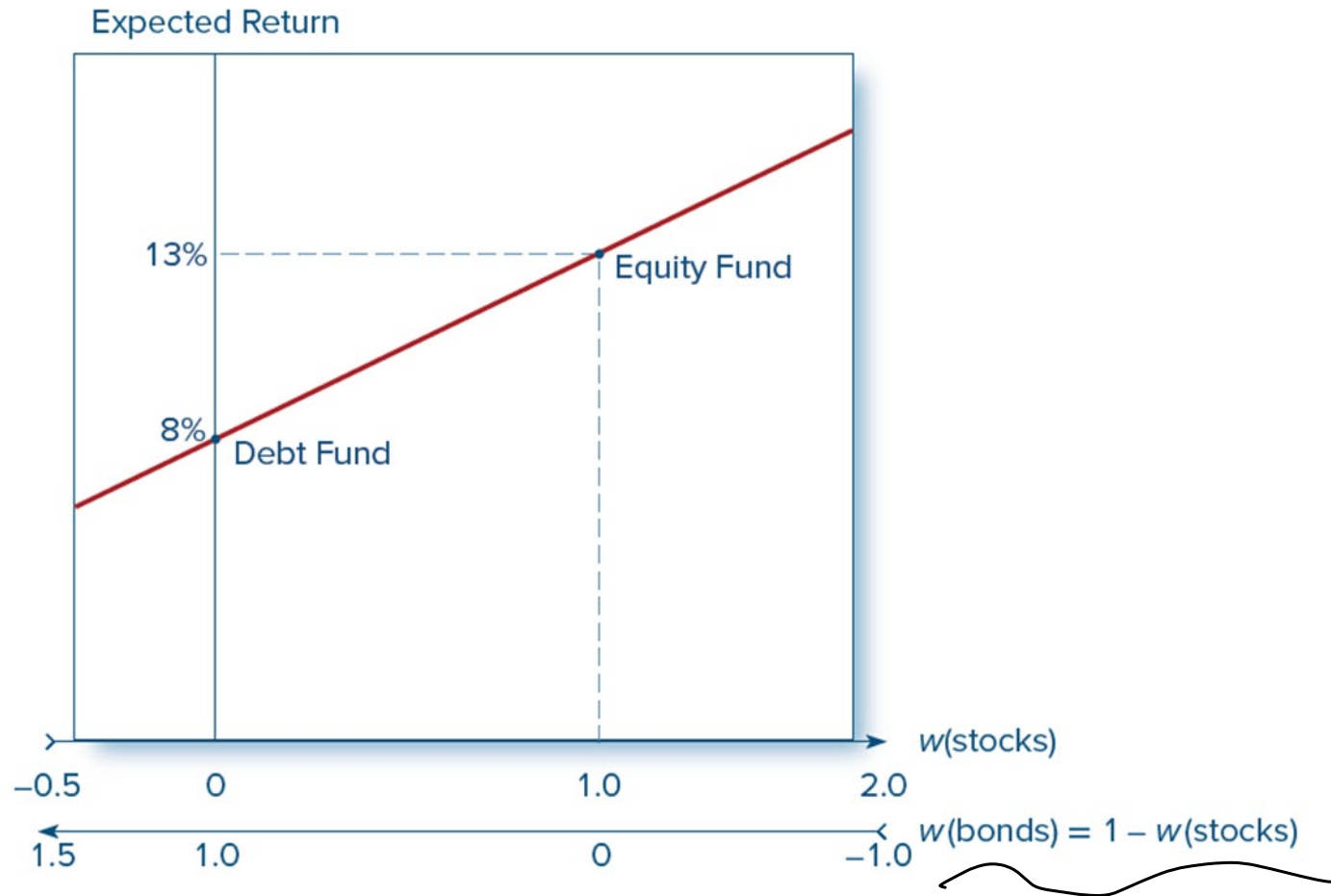
Variance:

$$\begin{aligned}
 \sigma_p^2 &= w_D^2 \sigma_D^2 + w_E^2 \sigma_E^2 + 2w_D w_E \text{Cov}(r_D, r_E) \\
 &= .50^2 \times 12\%^2 + .50^2 \times 20\%^2 + 2 \times .5 \times .5 \times 72 = 172
 \end{aligned}$$

**Standard
Deviation:**

$$\sigma_p = \sqrt{172} = 13.23\%$$

Figure 7.3 Portfolio Expected Return as a Function of Weight in Equity



Portfolio expected return as a function of investment proportions.

Table 7.2 Computation of Portfolio Variance From the Covariance Matrix

A. Bordered Covariance Matrix		
Portfolio Weights	w_D	w_E
w_D	$\text{Cov}(r_D, r_D)$	$\text{Cov}(r_D, r_E)$
w_E	$\text{Cov}(r_E, r_D)$	$\text{Cov}(r_E, r_E)$
B. Border-Multiplied Covariance Matrix		
Portfolio Weights	w_D	w_E
w_D	$w_D w_D \text{Cov}(r_D, r_D)$	$w_D w_E \text{Cov}(r_D, r_E)$
w_E	$w_E w_D \text{Cov}(r_E, r_D)$	$w_E w_E \text{Cov}(r_E, r_E)$
$w_D + w_E = 1$	$w_D w_D \text{Cov}(r_D, r_D) + w_E w_D \text{Cov}(r_E, r_D)$	$w_D w_E \text{Cov}(r_D, r_E) + w_E w_E \text{Cov}(r_E, r_E)$
Portfolio variance	$w_D w_D \text{Cov}(r_D, r_D) + w_E w_D \text{Cov}(r_E, r_D) + w_D w_E \text{Cov}(r_D, r_E) + w_E w_E \text{Cov}(r_E, r_E)$	

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Figure 7.4 Portfolio Standard Deviation as a Function of Weight in Equity

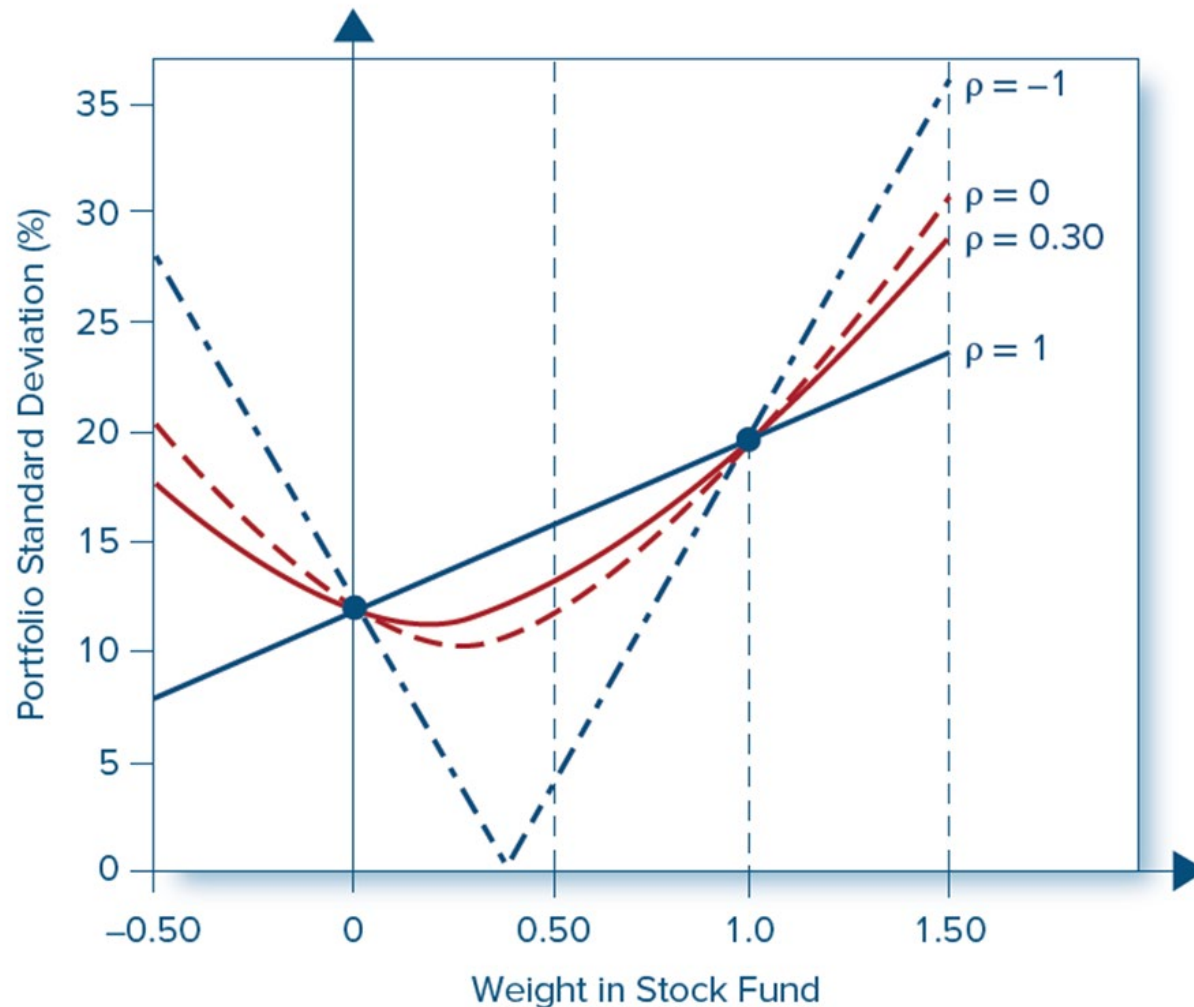


Figure 7.5 Portfolio Expected Return as a Function of Standard Deviation

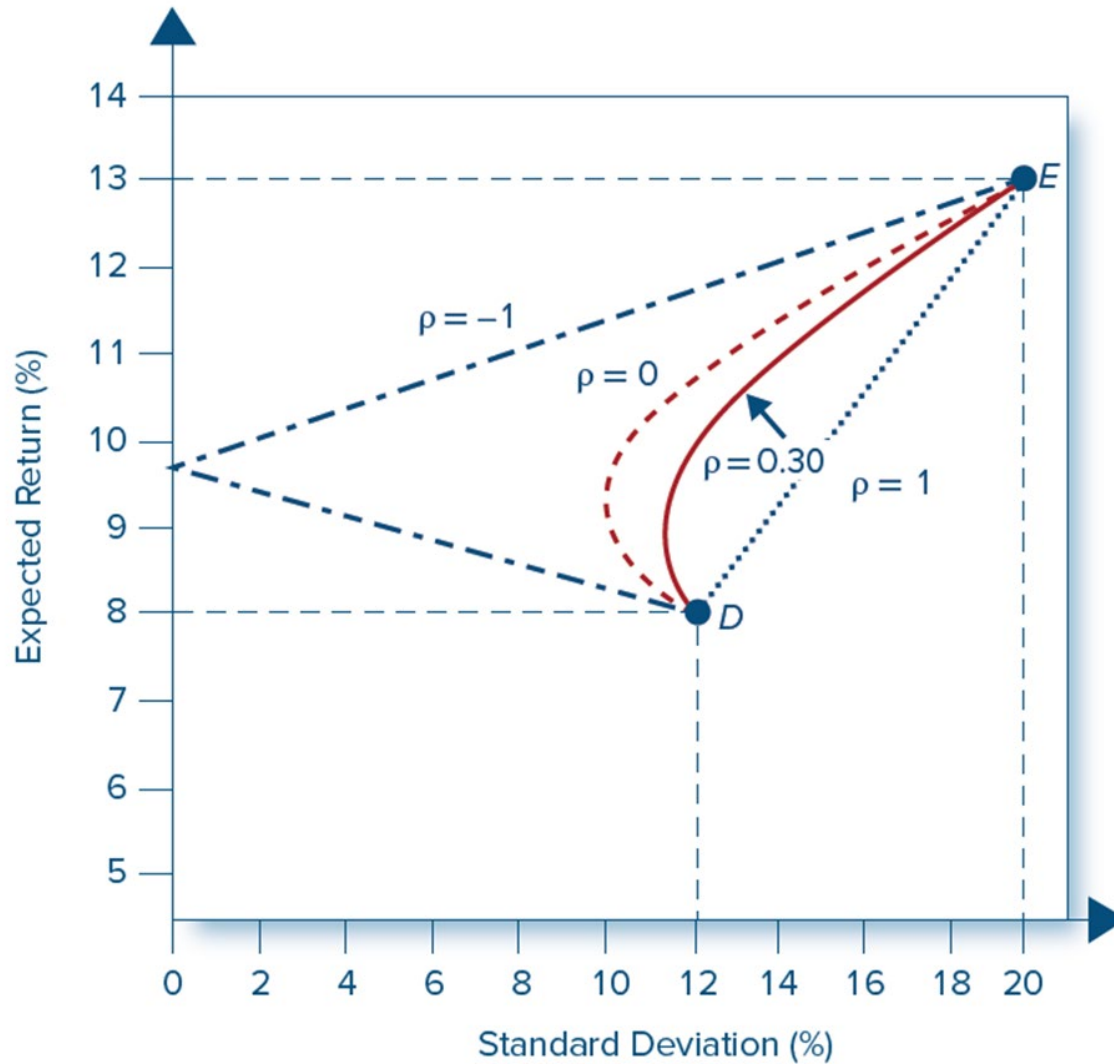


Table 7.3 Expected Return and Standard Deviation as a Function of Weights and Correlation Coefficients

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w_D	w_E	$E(r_p)$	Portfolio Standard Deviation for Given Correlation			
			$\rho = -1$	$\rho = 0$	$\rho = 0.30$	$\rho = 1$
0.00	1.00	13.00	20.00	20.00	20.00	20.00
0.10	0.90	12.50	16.80	18.04	18.40	19.20
0.20	0.80	12.00	13.60	16.18	16.88	18.40
0.30	0.70	11.50	10.40	14.46	15.47	17.60
0.40	0.60	11.00	7.20	12.92	14.20	16.80
0.50	0.50	10.50	4.00	11.66	13.11	16.00
0.60	0.40	10.00	0.80	10.76	12.26	15.20
0.70	0.30	9.50	2.40	10.32	11.70	14.40
0.80	0.20	9.00	5.60	10.40	11.45	13.60
0.90	0.10	8.50	8.80	10.98	11.56	12.80
1.00	0.00	8.00	12.00	12.00	12.00	12.00
Minimum Variance Portfolio						
	w_D		0.6250	0.7353	0.8200	—
	w_E		0.3750	0.2647	0.1800	—
	$E(r_p)$		9.8750	9.3235	8.9000	—
	σ_p		0.0000	10.2899	11.4473	—

Table 7.3

Expected return and standard deviation with various correlation coefficients

The Minimum-Variance Portfolio

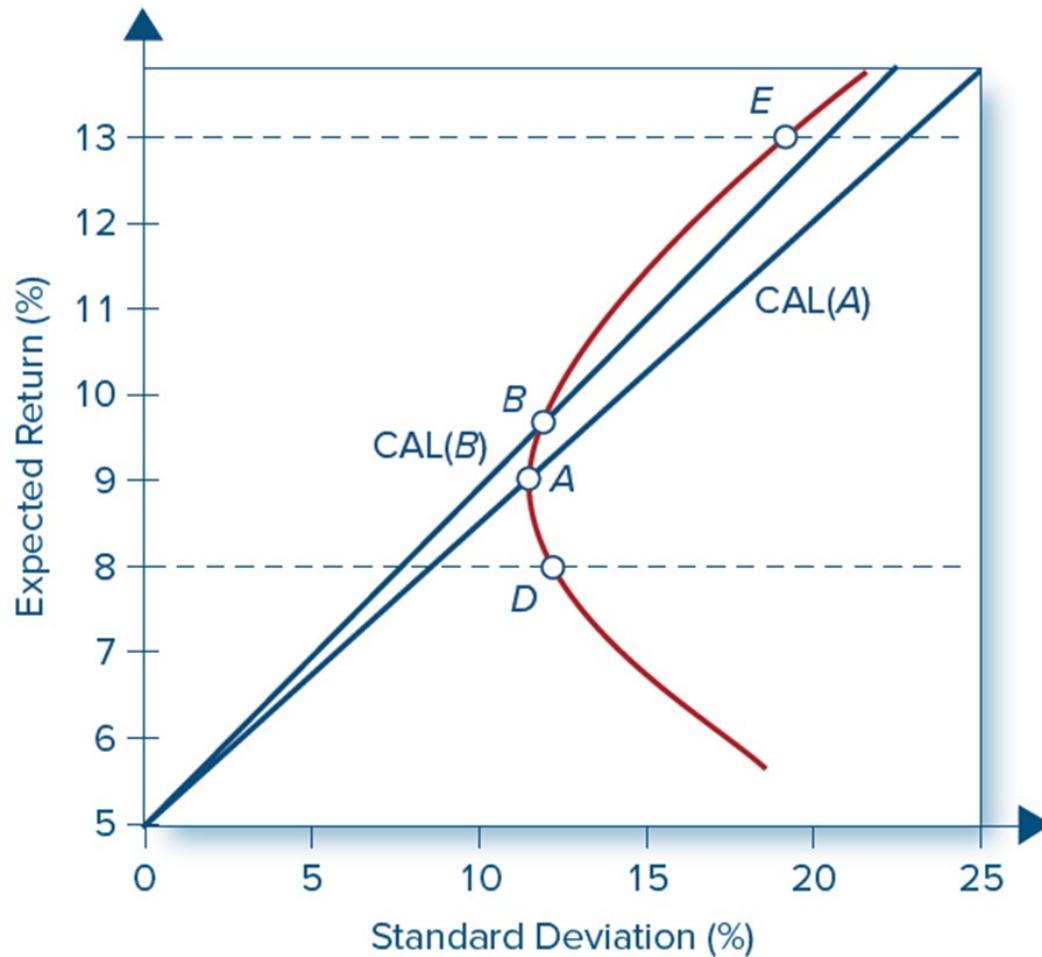
- The **minimum-variance portfolio** has a standard deviation *smaller than that of either of the individual component assets*.
- Risk reduction depends on the correlation:
 - If $\rho = +1.0$, no risk reduction is possible.
 - If $\rho = 0$, σ_p may be less than the standard deviation of either component asset.
 - If $\rho = -1.0$, a riskless hedge is possible.

w_D	w_E	$E(r_p)$	Portfolio Standard Deviation for Given Correlation			
			$\rho = -1$	$\rho = 0$	$\rho = 0.30$	$\rho = 1$
			Minimum Variance Portfolio			
		w_D	0.6250	0.7353	0.8200	—
		w_E	0.3750	0.2647	0.1800	—
		$E(r_p)$	9.8750	9.3235	8.9000	—
		σ_p	0.0000	10.2899	11.4473	—

Asset Allocation With Stocks, Bonds, and Bills

- Extending to include riskless asset (T-bills)
 - The optimal combination becomes linear
 - A single combination of risky and riskless assets will dominate
- Investors naturally want the risky portfolio that offers the highest (expected) reward-to-volatility ratio.
- The steepest capital allocation line (CAL) intersects with that best risky portfolio.

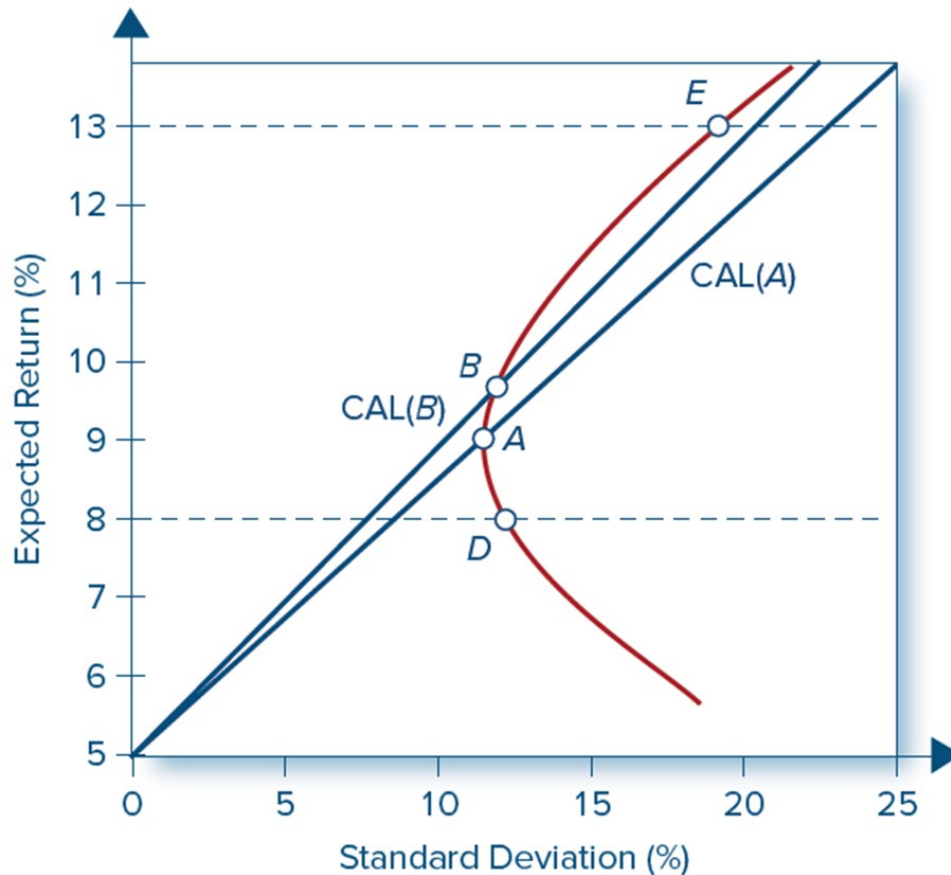
Figure 7.6 The Opportunity Set of the Debt and Equity Funds and Two Feasible CALs



Portfolio A
 $E(r_A) = 8.9\%$
 $\sigma_A = 11.45\%$

Portfolio B
 $E(r_B) = 9.5\%$
 $\sigma_B = 11.70\%$

The Sharpe Ratio: Example



Portfolio A

$$E(r_A) = 8.9\%$$

$$\sigma_A = 11.45\%$$

$$S_A = \frac{8.9\% - 5.0\%}{11.45\%} = .34$$

Portfolio B

$$E(r_B) = 9.5\%$$

$$\sigma_B = 11.70\%$$

$$S_B = \frac{9.5\% - 5.0\%}{11.70\%} = .38$$

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The Sharpe Ratio

- Objective is to find the weights w_D and w_E that result in the highest slope of the CAL.
- Thus, our objective function is the Sharpe ratio:

$$S_p = \frac{E(r_p) - r_f}{\sigma_p}$$

Optimal Risky Portfolio ¹

- The tangent portfolio (P) in Figure 7.7
- The solution to the maximization of the reward-to-variability ratio (the slope of CAL).

$$w_D = \frac{[E(r_D) - r_f] \sigma_E^2 - [E(r_E) - r_f] \sigma_D \sigma_E \rho_{DE}}{[E(r_D) - r_f] \sigma_E^2 + [E(r_E) - r_f] \sigma_D^2 - [E(r_D) - r_f + E(r_E) - r_f] \sigma_D \sigma_E \rho_{DE}}$$

$$w_E = 1 - w_D$$

Optimal Risky Portfolio ²

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Expected return, $E(r)$	8%	13%
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Covariance, $\text{Cov}(r_D, r_E)$	72	
Correlation coefficient, ρ_{DE}	0.30	

Risk-free rate 5%

$$w_D = \frac{[8 - 5]400 - [13 - 5]72}{[8 - 5]400 + [13 - 5]144 - [8 - 5 + 13 - 5]72} = .40$$

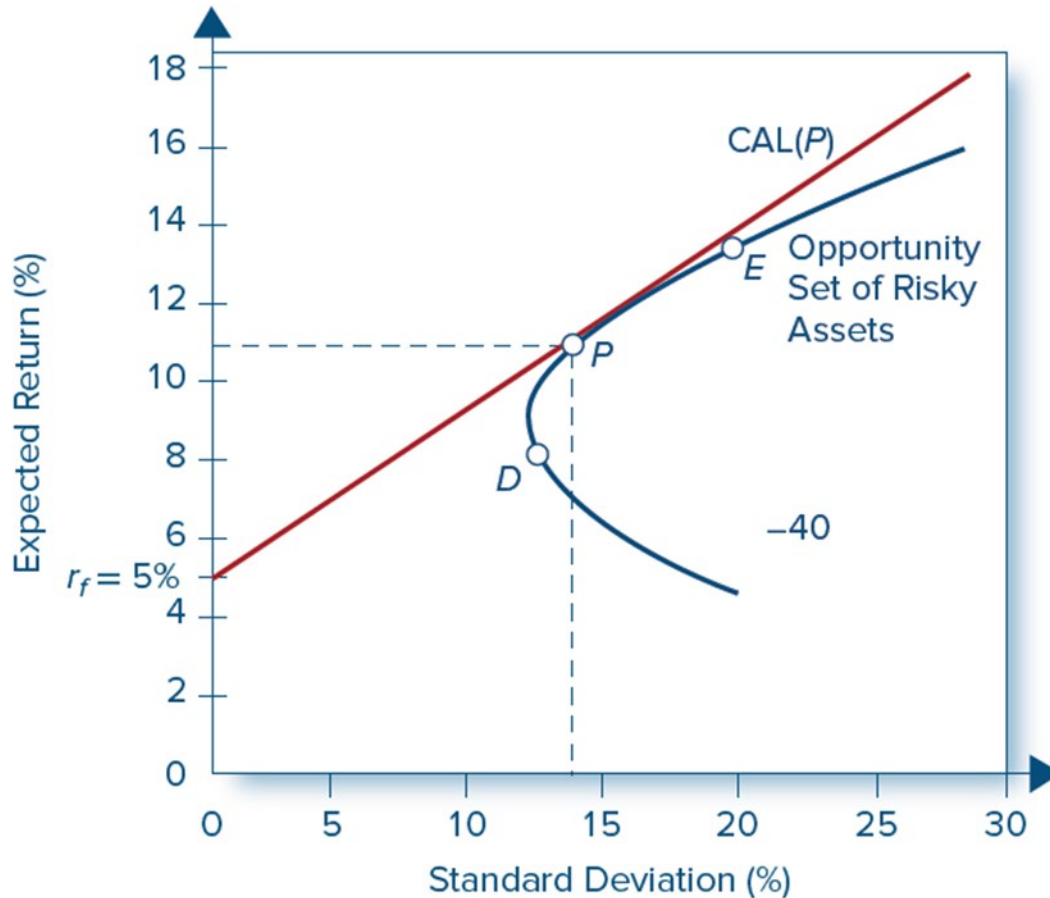
$$w_E = 1 - .40 = .60$$

The expected return and standard deviation of this optimal portfolio P

$$E(r_P) = (0.4 \times 8) + (0.6 \times 13) = 11\%$$

$$\sigma_P = [(.4^2 \times 144) + (.6^2 \times 400) + (2 \times .4 \times .6 \times 72)]^{1/2} = 14.2\%$$

Figure 7.7 Debt and Equity Funds With the Optimal Risky Portfolio



Optimal Risky Portfolio

$$E(r_P) = 11\%$$

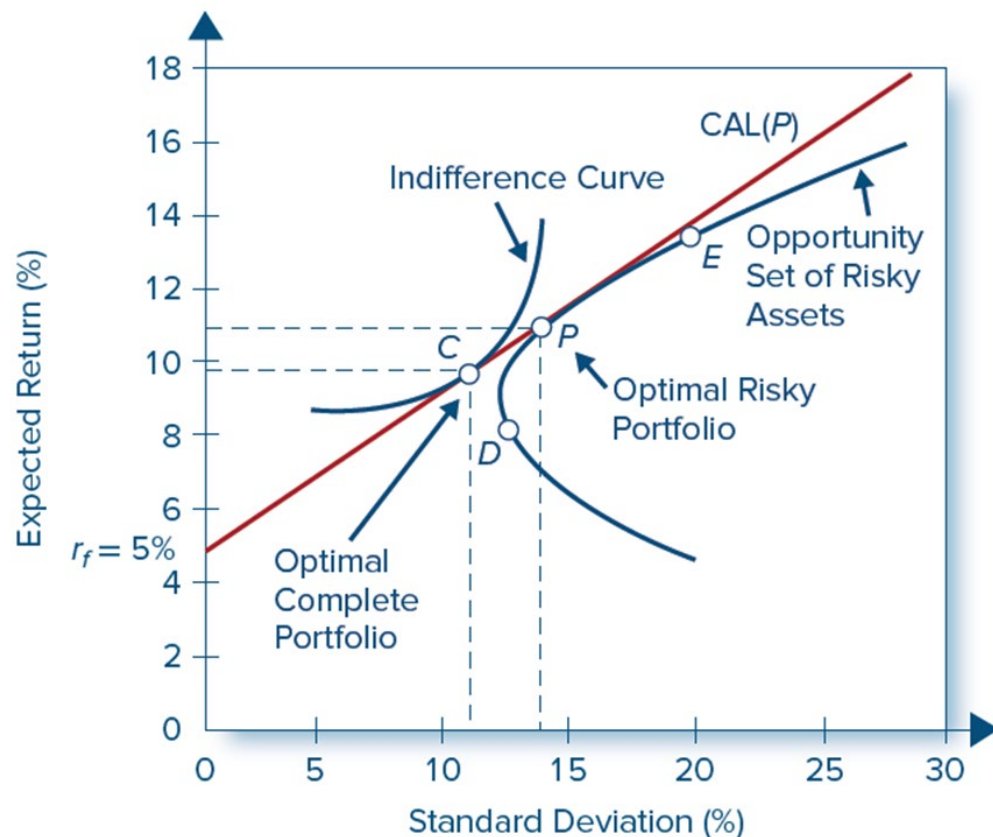
$$\sigma_P = 14.2\%$$

$$S_P = \frac{E(r_P) - r_f}{\sigma_P}$$

$$= \frac{11\% - 5\%}{14.2\%}$$

$$= .42$$

Figure 7.8 Determination of the Optimal Complete Portfolio



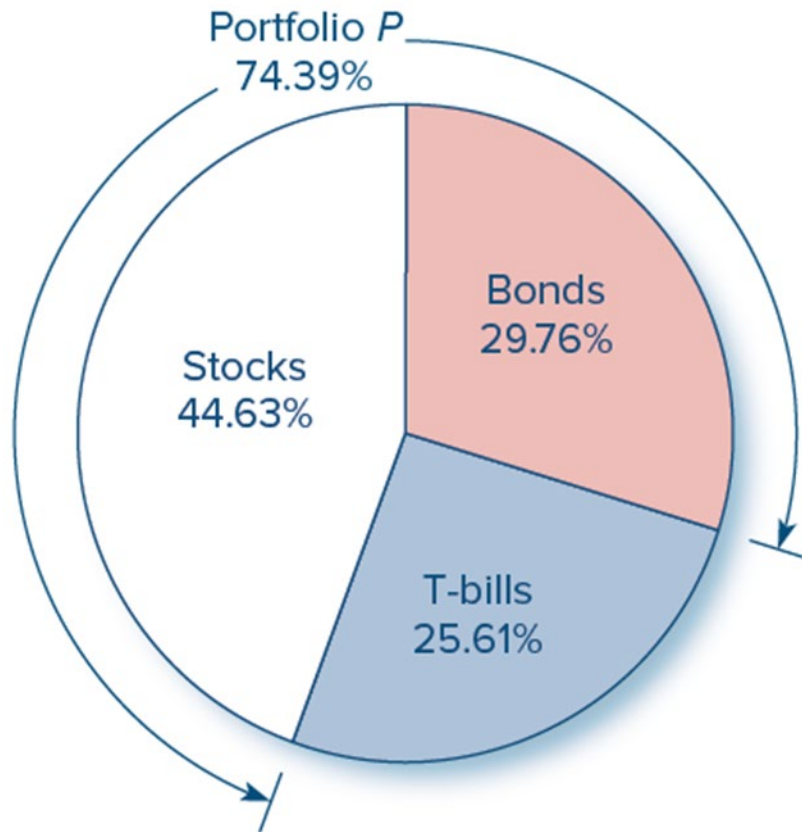
Optimal Allocation to P

$$A = 4$$

$$y = \frac{E(r_P) - r_f}{A\sigma_P^2}$$

$$= \frac{11\% - 5\%}{4 \times (14.2\%)^2} = .7439$$

The Proportions of the Optimal Complete Portfolio



Weight in bonds: $(40\%)(0.7439) = 29.76\%$
Weight in stocks: $(60\%)(0.7439) = 44.63\%$

Overall Portfolio

$$E(r_p) = 11\% \quad y = .7439$$

$$\sigma_p = 14.2\% \quad r_f = 5\%$$

$$\begin{aligned} E(r_{\text{Overall}}) &= y \times E(r_p) + (1 - y) \times r_f \\ &= .7439 \times 11\% + .2561 \times 5\% \\ &= 9.46\% \end{aligned}$$

$$\sigma_{\text{Overall}} = .7439 \times 14.2\% = 10.56\%$$

$$S_{\text{Overall}} = \frac{9.46\% - 5\%}{10.56\%} = .42$$

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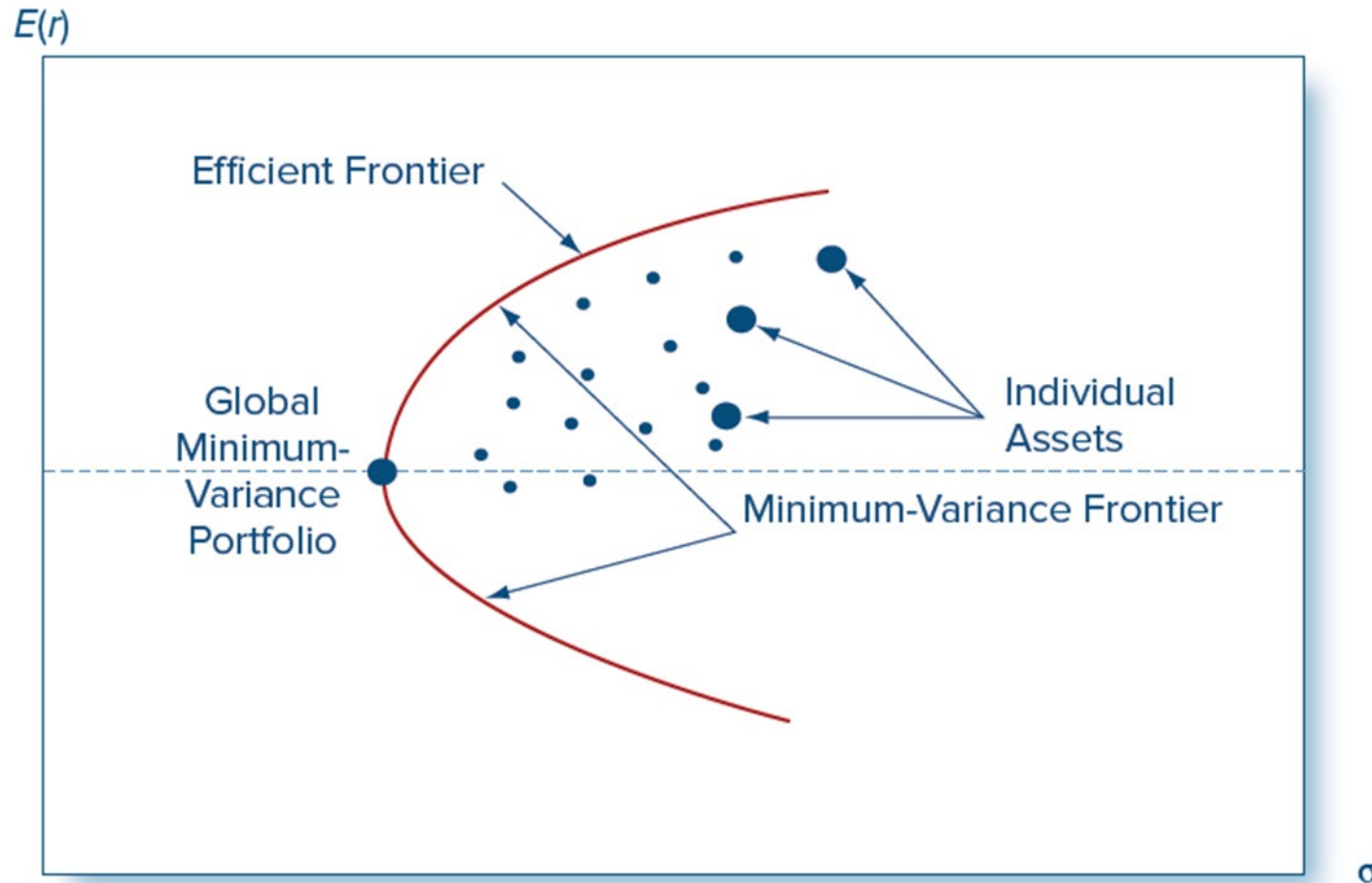
Markowitz Portfolio Optimization Model ¹

- Generalize the prior portfolio construction problem to the case of many risky assets and one risk-free asset
 - Step 1. Identify the risk-return combinations available from the set of risky assets
 - Step 2. Identify the optimal portfolio of risky assets by finding the portfolio weights that result in the steepest CAL
 - Step 3. Choose appropriate mix of risk-free asset and optimal risky portfolio to form complete portfolio

Markowitz Portfolio Optimization Model ²

- Minimum-variance frontier: A graph of the lowest possible variance that can be attained for a given portfolio expected returns.
- **Efficient frontier** of risky assets
- **Input list:** The estimates of the set of expected rates of return and a covariance matrix generated by the security analysts
- Separation property: 1. determination of optimal risky portfolio; 2. allocation of T-bills versus risky portfolio based on personal preference

Figure 7.10 The Minimum-Variance Frontier of Risky Assets



$$\sigma_P^2 = \sum_{i=1}^n \sum_{j=1}^n w_i w_j \text{Cov}(r_i, r_j)$$

Markowitz Portfolio Optimization Model ³

Security selection

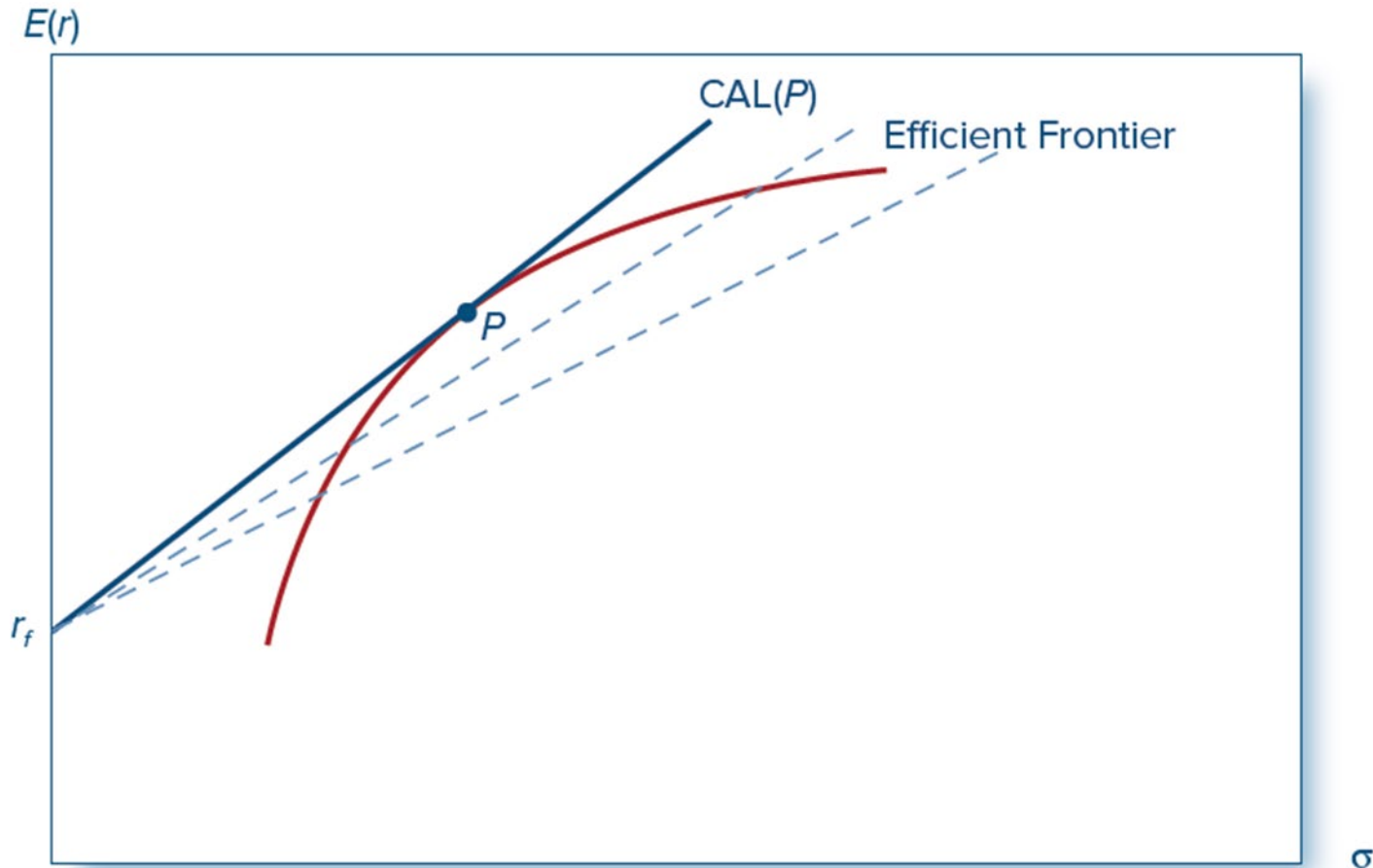
- Step 1: Determine the risk-return opportunities available
 - **Minimum-variance frontier** of risky assets
- All portfolios that lie on the minimum-variance frontier from the global minimum-variance portfolio and upward provide the best risk-return combinations
 - Efficient frontier of risky assets is the portion of the frontier that lies above the global minimum-variance portfolio

Markowitz Portfolio Optimization Model ⁴

Security selection (continued)

- Step 2: Search for the CAL with the highest Shape ratio (i.e., the steepest slope)
- Individual investor chooses the appropriate mix between the optimal risky portfolio P and T-bills
- Everyone invests in P , regardless of their degree of risk aversion
 - More risk averse investors put less in P
 - Less risk averse investors put more in P

Figure 7.11 The Efficient Frontier of Risky Assets With the Optimal CAL



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Markowitz Portfolio Optimization Model ⁵

Step 3: Capital allocation and the **separation property**

- Portfolio choice problem may be separated into two independent tasks
 - Determination of the optimal risky portfolio is purely technical
 - Allocation of the complete portfolio to risk-free versus the risky portfolio depends on personal preference

Markowitz Portfolio Optimization Model ⁶

- The Power of Diversification
 - Recall:

$$\sigma_p^2 = \sum_{i=1}^n \sum_{j=1}^n w_i w_j \text{Cov}(r_i, r_j)$$

- Assume we define the average variance and average covariance of the securities as:

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n \sigma_i^2$$

$$\overline{\text{Cov}} = \frac{1}{n(n-1)} \sum_{\substack{j=1 \\ j \neq i}}^n \sum_{i=1}^n \text{Cov}(r_i, r_j)$$

Markowitz Portfolio Optimization Model ⁷

- The power of diversification (continued).
- We can then express portfolio variance as

$$\begin{aligned}\sigma_P^2 &= \frac{1}{n} \sum_{i=1}^n \left(\frac{1}{n} \sigma_i^2 \right) + \sum_{\substack{j=1 \\ j \neq i}}^n \sum_{i=1}^n \left(\frac{1}{n^2} \text{Cov}(r_i, r_j) \right) \\ &= \frac{1}{n} \sigma^2 + \frac{n-1}{n} \overline{\text{Cov}} = \frac{1}{n} \sigma^2 + \left(1 - \frac{1}{n} \right) \overline{\text{Cov}} \quad (\#)\end{aligned}$$

- Although firm-specific risk, represented by the first term in the above equation (#), can be diversified away as n becomes larger, the second term simply approaches $\overline{\text{Cov}}$ as n becomes larger.

Markowitz Portfolio Optimization Model ⁸

- The power of diversification (continued).

$$\sigma_P^2 = \frac{1}{n} \bar{\sigma}^2 + \left(1 - \frac{1}{n}\right) \overline{Cov} \quad (\#)$$

- The risk of a highly diversified portfolio depends on the covariance of the returns of the component securities.
- Therefore, the irreducible risk of a diversified portfolio depends on the covariance of the returns of the component securities, which in turn is a function of the importance of systematic factors in the economy.
- This is because economy-wide risk factors impart positive correlation among stock returns.

Table 7.4 Risk Reduction of Equally Weighted Portfolios

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Universe Size n	Portfolio Weights $w = 1/n$ (%)	$\rho = 0$		$\rho = 0.40$	
		Standard Deviation (%)	Reduction in σ	Standard Deviation (%)	Reduction in σ
1	100	50.00	14.64	50.00	8.17
2	50	35.36		41.83	
5	20	22.36	1.95	36.06	0.70
6	16.67	20.41		35.36	
10	10	15.81	0.73	33.91	0.20
11	9.09	15.08		33.71	
20	5	11.18	0.27	32.79	0.06
21	4.76	10.91		32.73	
100	1	5.00	0.02	31.86	0.00
101	0.99	4.98		31.86	

Table 7.4

Risk reduction of
equally weighted port-
folios in correlated and
uncorrelated universes

For simplicity, assume common standard deviation and correlation coefficient
 $(0.4 \times 50 \times 50)^{1/2} = 31.62$

Markowitz Portfolio Optimization Model ⁹

- Optimal portfolios and non-normal returns.
- Fat-tailed distributions can result in extreme values of VaR and ES.
 - Practice way to estimate values of VaR and ES in the presence of fat tails is called *bootstrapping*.
- If other portfolios provide sufficiently better VaR and ES values than the mean–variance efficient portfolio, we may prefer these when faced with fat-tailed distributions.

Risk Pooling, Risk Sharing, and Time Diversification

- Risk pooling v.s. risk sharing.
- Variance of *average* insurance policy payoff decreases with the number of policies.

$$\text{Var}\left(\frac{1}{n} \sum_{i=1}^n x_i\right) = \frac{1}{n^2} \times n\sigma^2 = \frac{\sigma^2}{n}$$

- Variance of the *total* payoff becomes more uncertain.

$$\text{Var}\left(\sum_{i=1}^n x_i\right) = n\sigma^2$$

Risk Pooling and the Insurance Principle

- Risk pooling
 - Merging uncorrelated, risky projects as a means to reduce risk
 - It increases the *scale* of the risky investment by adding *additional* uncorrelated assets
- The insurance principle
 - Risk increases less than proportionally to the number of policies when the policies are uncorrelated
 - Sharpe ratio increases

Risk Pooling Example

- Investing \$1 in a single risky security A with risky rate of return r_A .
- Investing \$1 in another *uncorrelated* risky security B with risky rate of return r_B that has the same mean and variance.
 - The expected dollar profit on your \$2 investment is $2E(r)$
 - The variance of the dollar payoff is $\text{Var}(r_A + r_B) = 2\sigma^2$
- Clearly, this position is riskier than one invested only in A.

Risk Pooling Example

- Rate of return statistics of this example:

- The weights on each security are $\frac{1}{2}$

$$E(r_p) = \frac{1}{2} E(r) + \frac{1}{2} E(r) = E(r)$$

$$Var(r_p) = (\frac{1}{2})^2 \sigma^2 + (\frac{1}{2})^2 \sigma^2 + 2(\frac{1}{2})(\frac{1}{2})Cov(r_A, r_B) = \frac{1}{2} \sigma^2$$

$$SD(r_p) = \sigma_p = \sqrt{\frac{1}{2}} \sigma$$

- While its rate of return is more predictable ($\frac{1}{2} \sigma^2$), the dollar variance ($2 \sigma^2$) of the “risk-pooled portfolio” is double the variance of the one-asset portfolio.
- Risk pooling improves risk-return trade-off!

$$S_p = \frac{\text{Expected dollar risk premium}}{SD(\text{dollar profit})} = \frac{2[E(r) - r_f]}{\sigma\sqrt{2}} = \sqrt{2}S_A$$

Risk Sharing

- As risky assets are added, a portion of the pool is sold to maintain a risky portfolio of fixed size
- Risk sharing combined with risk pooling is the *key to the insurance industry*
- True diversification means spreading a portfolio of *fixed size across many assets*, not merely adding more risky bets

Risk Sharing Example

- A fixed \$1 investment : in security A \$ $\frac{1}{2}$ and in security B \$ $\frac{1}{2}$
- The expected dollar profit on your \$1 investment is $E(r)$
- The variance of the dollar payoff is
$$\text{Var}(r_A + r_B) = \frac{1}{4} \sigma^2 + \frac{1}{4} \sigma^2 = \frac{1}{2} \sigma^2$$
- The Sharpe ratio for the risk-sharing portfolio is also $\sqrt{2}$ of the one-asset portfolio:

$$\frac{\text{Expected dollar risk premium}}{SD(\text{dollar profit})} = \frac{E(r) - r_f}{\sigma / \sqrt{2}} = \sqrt{2} S_A$$

Risk Pooling and Risk Sharing

- Risk pooling : Investing \$1 in A and \$1 in B. The gambler in Las Vegas who goes to the roulette table twice than once.
 - With every trip to the roulette table, percentage gains (in this case, losses), are more predictable, but dollar gains or losses are less so.
- Risk sharing : Share 100 bets with 100 friends. A 1/100 share of 100 bets is in fact less risky than on bet.

Time Diversification

- True diversification.
 - Requires holding fixed the total funds put at risk and spreading the exposure across multiple sources of uncertainty.

Example presented in Table 7.5.

	Investment Horizon (years)			Comment
	1	10	30	
1. Mean of average return	0.050	0.050	0.050	$= .05$
2. Mean of total return	0.050	0.500	1.500	$= .05 \cdot T$
3. Standard deviation of total return	0.300	0.949	1.643	$= .30 \sqrt{T}$
4. Standard deviation of average return	0.300	0.095	0.055	$= .30 / \sqrt{T}$
5. Prob(return > 0)	0.566	0.701	0.819	From normal distribution
6. 1% VaR total return	-0.648	-1.707	-2.323	Continuously compounded cumulative return
7. Cumulative loss at 1% VaR	0.477	0.819	0.902	$= 1 - \exp(\text{cumulative return from line 6})$
8. 0.1% VaR total return	-0.877	-2.432	-3.578	Continuously compounded return
9. Cumulative loss at 0.1% VaR	0.584	0.912	0.972	$= 1 - \exp(\text{cumulative return from line 8})$